



# CAMARA Famakan

Data Engineering · Machine Learning · Optimization

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## Profile

Final-year engineering student in Computer Science & Data Engineering at ENSA Khouribga. Versatile profile at the intersection of data, optimization, and software development: data pipelines, predictive modeling, combinatorial optimization formulation, and integrating generative AI into real applications. Certified **Associate Data Analyst** and **Associate Data Engineer** (DataCamp). Seeking a **final-year internship (PFE, 4 to 6 months)**.

## Experience

### PFA Intern — MaritimeFlow, Maritime Fleet Optimization

Aug – Sep 2026

[OCP Group](#) · [Khouribga](#)

- Built a full pipeline for cleaning and structuring geospatial (AIS) data, with clustering for port detection and construction of an origin-destination matrix.
- Trained a predictive model (LightGBM) for travel-time estimation, then formulated a combinatorial optimization problem (MILP) combined with a metaheuristic (ALNS), validated through discrete-event simulation.

### Initiation Internship — Data Engineering & Applications

Jul – Aug 2025

[OCP Group](#) · [Khouribga](#)

- Designed a normalized PostgreSQL schema and built a full-stack application (Node.js REST API, React interface) with Power BI dashboards for KPI monitoring in an industrial environment.

## Projects

### xAgriScan — Computer Vision + LLM

2025

[ResNet-50](#) · [Gemini](#) · [Flask](#) · [React](#)

- Application combining a vision model with an LLM to turn a technical diagnosis into a clear recommendation for a non-expert user.

### TSP Solver — Optimization Metaheuristics

2025

[Python](#) · [Flask](#) · [Simulated Annealing](#) · [VNS](#)

- Implemented and compared multiple optimization heuristics (VNS, Simulated Annealing, Greedy) for the Traveling Salesman Problem.

### Sales Management System — RAG Chatbot

2026

[Spring Boot](#) · [React](#) · [SQL](#) · [RAG](#)

- Full-stack sales management application with a RAG chatbot enabling natural-language data querying.

### Hadoop Cluster — MovieLens 32M

2025

[Hadoop](#) · [HDFS](#) · [MapReduce](#)

- Set up a multi-machine Hadoop cluster and implemented MapReduce jobs to process large-scale data, in a team setting.

## Education

### Engineering Cycle — Computer Science & Data Engineering

2024 – 2027

[ENSA Khouribga \(École Nationale des Sciences Appliquées\)](#)

### Integrated Preparatory Cycle

2021 – 2024

[ENSA Khouribga](#)

### Baccalaureate, Mathematical Sciences — Distinction

2021

[Lycée Sokona Drámara Yangué — Kéniéba, Mali](#)

## Skills

### Data & Optimization

- SQL · PostgreSQL · MongoDB · Power BI
- Python (Pandas, NumPy) · Machine Learning
- MILP · ALNS · Simulated Annealing · VNS · SimPy
- Big Data: Hadoop, Spark/PySpark

### Development & AI

- Java · Spring Boot · Flask · Node.js · React
- Docker · Git/Linux
- RAG, FAISS, LLM APIs, fine-tuning (BERT, AraT5v2, ResNet-50)

## Certifications

**DataCamp Certified** — Associate Data Analyst (Aug 2026) · Associate Data Engineer (Aug 2026) · AI Engineer for Data Scientists Associate (Aug 2026)

## Languages

Bambara — Native

French — Fluent

English — B2